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funkyheatmap: Visualising data frames with mixed data types

Robrecht Cannoodt* [1,2,3], **Louise Deconinck*** [1,2], **Artuur Couckuyt*** [1,2], **Nikolay S. Markov*** [6], **Luke Zappia** [3,7,9], **Malte D. Luecken** [7,9], **Marta Interlandi** [5], **Yvan Saeys†** [1,2], and **Wouter Saelens†** [4,10]

1 Data Mining and Modelling for Biomedicine group, VIB Center for Inflammation Research, Ghent, Belgium **2** Department of Applied Mathematics, Computer Science, and Statistics, Ghent University, Belgium **3** Data Intuitive, Lebbeke, Belgium **4** Laboratory of Myeloid Cell Biology in tissue homeostasis and regeneration, VIB Center for Inflammation Research, Ghent, Belgium **5** Institute of Medical Informatics, University of Muenster, Muenster, Germany **6** Division of Pulmonary and Critical Care Medicine, Northwestern University, Chicago, USA **7** Institute of Computational Biology, Computational Health Center, Helmholtz Munich, Munich, Germany **8** Department of Mathematics, School of Computation, Information and Technology, Technical University of Munich, Munich, Germany **9** Institute of Lung Health and Immunity (LHI), Helmholtz Munich, Comprehensive Pneumology Center (CPC-M), Germany; Member of the German Center for Lung Research (DZL) **10** Department of Biomedical Molecular Biology, Ghent University, Ghent, Belgium

*: Shared first authors, †: Shared last authors.

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- @rhenkin
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Summary

The {funkyheatmap} package offers a flexible and user-friendly solution for visualising data frames containing a mixture of categorical, proportional, and text-based data. It simplifies the creation of informative and visually appealing heatmaps while providing extensive customization options to tailor the output. This tool is especially valuable in research settings for summarising and communicating complex results, such as those encountered in benchmarking studies.

The package is available on CRAN and PyPI and has a JavaScript port in development. For a showcase of examples and an overview of the different implementations, visit the project website funkyheatmap.github.io.

Statement of need

Data visualisation is fundamental to exploratory data analysis and communicating findings. While powerful tools like ggplot2 (Wickham, 2009), Matplotlib (Hunter, 2007) and D3.js (Bostock et al., 2011) exist, they often require complex scripting to generate comprehensive visualisations for data frames containing a mix of data types. {funkyheatmap} addresses this challenge by:

- **Seamless Handling of Mixed Data**: Automates the selection of appropriate visualisations (rectangles, bars, pie charts, text) based on data types.
- **Customization**: Provides granular control over colours, groupings, geometries, and annotations for tailored results.
- **Accessibility**: Offers a simplified interface for basic use and detailed documentation for advanced customization.

{funkyheatmap} has proven its utility in benchmarking studies within single-cell omics (Li et al., 2023; Luecken et al., 2022, 2024; Saelens et al., 2019; Sang-Aram et al., 2024; Yan & Sun,

2023), but its applications extend to diverse fields where visualisation of mixed data types is needed.

Functionality

Figure 1 showcases the functionality of {funkyheatmap}, namely:

- **Diverse Geometries**: Supports a range of geometries (rectangles, bars, pie charts, text, images) to effectively represent different data types.
- **Hierarchical Categorical Grouping**: Facilitates the organisation of rows and columns into semantic groups with distinct colour palettes.

Besides this, the package includes comprehensive documentation, vignettes, and a test suite.

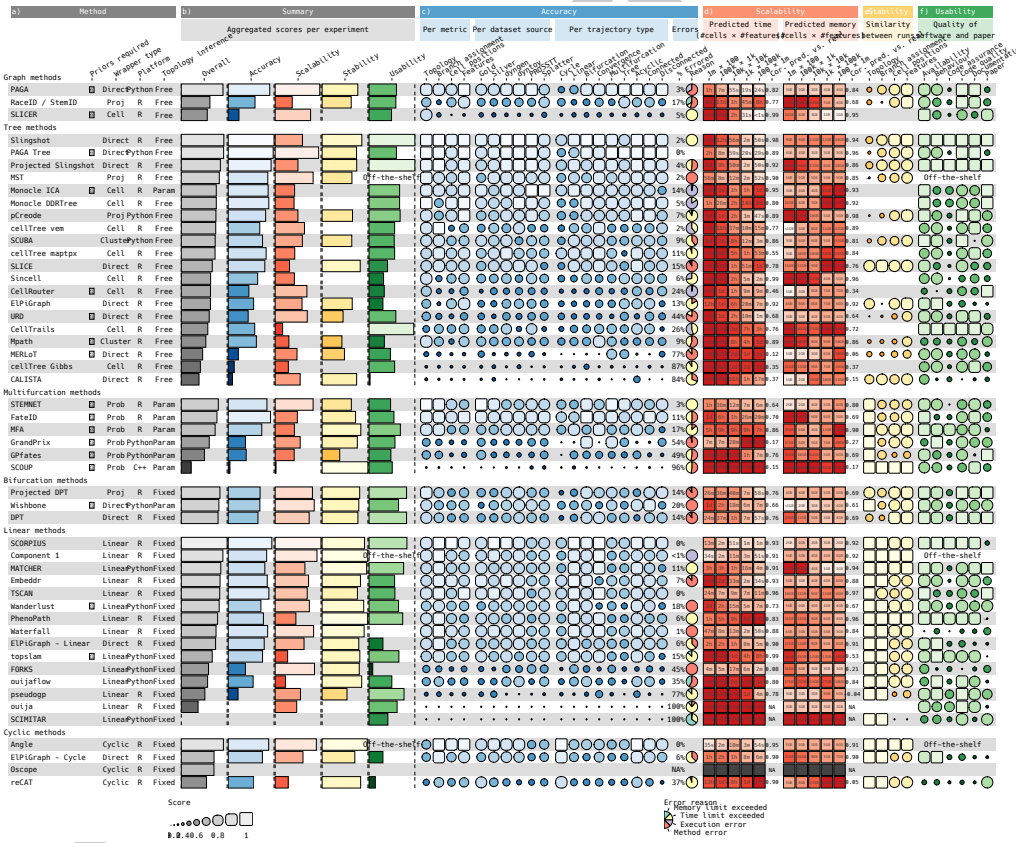

Figure 1: An example of a {funkyheatmap} visualisation using data from a benchmarking study of trajectory inference methods (Saelens et al., 2019).

See the following Table for more information regarding the recommended geom for different types of data.

Data type	Example	Recommended geom
Numerical data	Scores from 0 to 1	funkyrect
Aggregated data	The mean of scores	bar
Measurement data	3MB or 4h	rect + text overlay
Categorical data	R or Python	text or image
Proportional data	80% success, 10% OOM, 10% failed	pie

Recommended geometries in {funkyheatmap} for different data types. The table presents the suggested visualisation methods (geoms) based on the data type of the columns. These recommendations provide a starting point for users to select the most appropriate visual representation for their specific data.

Conclusion

{funkyheatmap} streamlines the creation of publication-quality visualisation for mixed data types, empowering researchers and data scientists to communicate their results effectively. The development of funkyheatmappy (Python) and funkyheatmapjs (JavaScript) will further expand the accessibility and functionality of this visualisation solution.

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